

Sharp Energy Selection Before Residual Grouping: Capacity Saturation, Cofactor Occupancy, and the Exact Coverage Threshold

Liang Wang

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

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Abstract

The preceding dictionary of literal factors isolates a positive-volume family of prime-squarefree source cells on which the sampled orbit profile has a uniform pointwise overlap with its carrier. That geometric statement does not by itself select a positive fraction of the actual-coefficient-weighted carrier energy: a bounded inherited coefficient may vanish on the compatible cells, and the number of retained cofactors may concentrate on their complement. We solve the finite-atom selection problem exactly, before any coherent prime sum or residual grouping is performed.

Let a good set have structural mass δ , let $|\zeta_m| \leq C$, and let κ be the coefficient-weighted energy divided by the total capacity C^2W . If σ is the fraction of that energy on the good set, then

$$\sigma \geq \left(1 - \frac{1 - \delta}{\kappa}\right)_+.$$

The threshold $\kappa > 1 - \delta$, the inequality, and all endpoint cases are sharp. We also obtain the sharp bound in the presence of a good-cell floor $|\zeta_m| \geq \theta C$, including the optimal combination of the local floor and the global capacity certificate.

Structural density is transported through source shapes, sampled-carrier weights, and cofactor occupancy by the fractional-linear map

$$T_\lambda(u) = \frac{\lambda u}{\lambda u + 1 - u}.$$

Its odds multiply exactly, so successive distortion exponents add. This gives an explicit coverage ledger for the physical fixed- h_0 packet and identifies three sufficient routes: a coefficient floor plus balanced occupancy, near-saturation of total coefficient capacity, or a finite compatible cover of all active carrier energy. None is supplied for the full inherited coefficient class by an upper bound alone.

Finally, arbitrary diagonal phases leave every pre-terminal selection quantity unchanged, while a two-atom collision in one full post-prime output fiber—whose coordinates include $s = d(m)r$ —already creates a nontrivial kernel. Thus phase distribution cannot repair a magnitude deficit, and positive atomic coverage cannot be pushed through terminal grouping without a new coherence estimate. The results close a sharp L0 selection calculation and give conditional L1 specializations; they prove no post-grouping lower bound, parity improvement, or twin-prime statement.

Keywords: energy selection; capacity saturation; cofactor occupancy; prime-squarefree cells; residual grouping; sieve parity.

Convention. Every atomic energy in the selection statements is the actual-coefficient-weighted carrier energy on retained pre-terminal labels. The unseparated physical-profile norm and a grouped output are named separately. The shift $h_0 \neq 0$ is fixed. Unit phases are never converted into magnitude information, and the many-to-one grouping $s = d(m)r$ is never moved through a squared norm.

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1 Introduction

The TPC reduction studied in [1–3] contains several logically different kinds of “mass.” A fixed prime-squarefree source rectangle can contain a positive proportion of the admissible rows. Fixed smooth source shapes can give those rows a positive proportion of a coefficient-free base energy. The sampled carrier and repetition over retained cofactors then change the row weights. Finally, the inherited coefficient ζ_m may redistribute, or completely remove, the carrier energy. Conflating any two of these statements turns a support theorem into an unjustified lower bound.

TPC–52 [4] proved a conditional pre-terminal profile estimate once a fraction σ_X of the actual-coefficient carrier energy lies on compatible source fibers. TPC–53 [5] supplied the literal factor dictionary, exact atomic normalization, and a fixed compatible box on which the sampled profile-to-carrier ratio is uniformly positive. It also exhibited the remaining obstruction: the inherited theory assumes only $|\zeta_m| \ll 1$, and does not force the actual-coefficient carrier energy or the cofactor multiplicity to visit that box. The present paper asks the narrow next question:

Given the structural size of a compatible set and only specified information about coefficient capacity and cofactor occupancy, what is the best possible lower bound for its actual-coefficient pre-terminal carrier energy?

This question is finite-dimensional. Its answer is therefore exact, not asymptotic. The arithmetic work is to identify which weights occur in the physical packet and which hypotheses the inherited papers actually provide.

1.1 Main results

Our first result is a sharp capacity law. Let W be total structural mass, let the good set carry mass δW , and suppose $|\zeta_m| \leq C$. If the coefficient energy is $\kappa C^2 W$, the bad set can hold at most $(1 - \delta)C^2 W$. Hence the good energy fraction obeys

$$\sigma \geq \frac{(\kappa - (1 - \delta))_+}{\kappa}.$$

Both regimes are attained: below the threshold one may place all energy on bad atoms, while above it one first saturates every bad atom. In terms of the unsaturated capacity $\epsilon = 1 - \kappa$, positive selection is forced exactly when $\epsilon < \delta$.

The second result describes changes of measure sharply. If a multiplier is at least n_- on good atoms and at most n_+ on bad atoms, then a good fraction u becomes at least

$$\mathsf{T}_{n_-/n_+}(u) = \frac{n_- u}{n_- u + n_+(1 - u)}.$$

The identity

$$\frac{\mathsf{T}_\lambda(u)}{1 - \mathsf{T}_\lambda(u)} = \lambda \frac{u}{1 - u}$$

shows why independent losses multiply in odds coordinates. Source-shape, sampled-carrier, cofactor-occupancy, and coefficient-floor losses therefore compose without the slack introduced by repeatedly discarding denominators.

Our third result is negative but operationally decisive. All selection quantities before the terminal map depend only on $|\zeta_m|$, so rowwise unit phases cannot change them. After the terminal map, two admissible inputs can collide only when their full output data agree; one coordinate is $s = d(m)r$. Any such output fiber containing two nonzero inputs has a one-dimensional cancellation direction. A pre-terminal lower bound therefore does not survive grouping merely by positivity; one needs new information adapted to the distinguished arithmetic coefficients.

1.2 What is and is not arithmetic progress

The sharp inequalities are unconditional L0 statements. Their specialization to the literal fixed- h_0 carrier weights is also exact. The earlier source-cell and profile-overlap certificates then give an L1 route under any one of several explicit additional hypotheses. The full inherited class currently supplies only a coefficient upper bound and an arbitrary supported cofactor set. It supplies neither a nonvanishing coefficient floor, a near-capacity estimate, nor balanced cofactor occupancy. Consequently we do *not* claim a positive coverage constant for the full inherited packet.

Nothing here controls the coherent sum over the terminal prime, proves stability of residual grouping, identifies the residual Gram matrix with the native affine Gram matrix, or performs Fourier localization at a moving shift. Only the literal fixed shift h_0 is physical. In particular, the paper contains no parity-breaking estimate and no twin-prime consequence.

1.3 Organization

[Section 2](#) defines the selection variables and the exact odds calculus. [Section 3](#) proves the capacity and coefficient floor theorems. [Section 4](#) treats reweighting, multiple stages, and a finite-cover alternative. [Section 5](#) inserts the literal prime-squarefree weights. [Section 6](#) proves the phase-invariance and grouping obstructions. [Section 7](#) records the polynomial budget, claim level, and the next gate.

2 The pre-terminal coverage ladder

2.1 Finite weighted atoms

Let $\mathcal{M}$ be a nonempty finite set, let $w_m \geq 0$, and write

$$W(S) := \sum_{m \in S} w_m, \quad W := W(\mathcal{M}) > 0.$$

Atoms of zero weight may be deleted, so no result changes if one assumes $w_m > 0$. Fix a good set $\mathcal{G} \subseteq \mathcal{M}$ and define its structural fraction by

$$\delta := \frac{W(\mathcal{G})}{W} \in [0, 1]. \tag{1}$$

For coefficients $\zeta_m \in \mathbb{C}$, put

$$E(S) := \sum_{m \in S} w_m |\zeta_m|^2, \quad E := E(\mathcal{M}). \tag{2}$$

Whenever $E > 0$, the actual good-energy fraction is

$$\sigma := \frac{E(\mathcal{G})}{E}. \tag{3}$$

These are atomic squared norms. No sum of complex amplitudes has yet occurred.

If $|\zeta_m| \leq C$ for a fixed $C > 0$, the total available capacity is C^2W . We normalize the occupied capacity by

$$\kappa := \frac{E}{C^2W} \in (0, 1]. \quad (4)$$

The three numbers have different meanings: δ is structural, κ is global magnitude information, and σ is the desired localized coefficient-energy fraction.

2.2 The odds transform

For $\lambda \geq 0$ and $0 \leq u \leq 1$, set

$$\mathsf{T}_\lambda(u) := \frac{\lambda u}{\lambda u + 1 - u}, \quad (5)$$

with the endpoint convention $\mathsf{T}_0(1) := 1$; thus $\mathsf{T}_0(u) = 0$ for $u < 1$. There is no joint continuous extension at the exceptional corner $(0, 1)$; the convention is only bookkeeping and will not be used with a zero lower multiplier.

Lemma 2.1 (Odds calculus). *For $0 < u < 1$ and $\lambda > 0$,*

$$\frac{\mathsf{T}_\lambda(u)}{1 - \mathsf{T}_\lambda(u)} = \lambda \frac{u}{1 - u}. \quad (6)$$

Moreover, T_λ is increasing in both variables and

$$\mathsf{T}_\lambda(\mathsf{T}_\mu(u)) = \mathsf{T}_{\lambda\mu}(u) \quad (7)$$

whenever the displayed expressions are defined by the stated endpoint conventions.

Proof. Both identities follow by direct substitution. Differentiation gives

$$\partial_u \mathsf{T}_\lambda(u) = \frac{\lambda}{(1 - (1 - \lambda)u)^2}, \quad \partial_\lambda \mathsf{T}_\lambda(u) = \frac{u(1 - u)}{(1 - (1 - \lambda)u)^2},$$

in the interior, proving monotonicity. The ordinary endpoints follow by one-variable limits, while the exceptional corner and the composition law there are checked directly from the convention. $\square$

The transform is the exact bookkeeping device for the rest of the paper. If a change of weight decreases the good-to-bad odds by at most a factor λ , it changes the good fraction by T_λ . A chain of changes multiplies the odds factors, rather than adding an artificial loss at each denominator.

2.3 Five distinct levels

For later reference we separate the following levels.

- (C1) *Geometric occupancy*: counting or prime-squarefree base mass on a compatible source box.
- (C2) *Source base energy*: geometric mass reweighted by fixed logarithmic and smooth source shapes, but not by ζ_m .
- (C3) *Sampled carrier energy*: source base energy reweighted by the coefficient-free sampled carrier.
- (C4) *Lifted carrier energy*: sampled carrier energy repeated over the actual retained cofactors.
- (C5) *Actual-coefficient carrier energy*: lifted carrier energy multiplied by $|\zeta_m|^2$.

Levels C1–C5 are all pre-terminal. The coherent prime sum and the map grouping $s = d(m)r$ lie after this ladder. Thus positivity permits exact selection within the ladder, but not across the terminal boundary.

3 Sharp selection from coefficient capacity

We first keep the structural weights fixed and ask what a coefficient upper bound plus one scalar global energy statistic can force.

Theorem 3.1 (Sharp capacity threshold). *Let $0 \leq \delta \leq 1$, let $|\zeta_m| \leq C$, and suppose $E > 0$. Then*

$$\sigma \geq \frac{(\kappa - (1 - \delta))_+}{\kappa} = \left(1 - \frac{1 - \delta}{\kappa}\right)_+. \quad (8)$$

For every admissible pair (δ, κ) , the bound is attained by coefficients whose squared modulus is constant separately on $\mathcal{G}$ and $\mathcal{M} \setminus \mathcal{G}$. In particular, global information forces $\sigma > 0$ if and only if

$$\kappa > 1 - \delta. \quad (9)$$

Proof. The bad set has capacity

$$E(\mathcal{M} \setminus \mathcal{G}) \leq C^2 W(\mathcal{M} \setminus \mathcal{G}) = C^2 W(1 - \delta).$$

Since $E = \kappa C^2 W$, subtraction gives

$$E(\mathcal{G}) \geq C^2 W(\kappa - (1 - \delta)).$$

Combining this with $E(\mathcal{G}) \geq 0$ and dividing by E proves (8).

Assume first $0 < \delta < 1$. If $0 < \kappa \leq 1 - \delta$, set

$$|\zeta_m|^2 = 0 \quad (m \in \mathcal{G}), \quad |\zeta_m|^2 = C^2 \frac{\kappa}{1 - \delta} \quad (m \notin \mathcal{G}).$$

Then $E = \kappa C^2 W$ and $\sigma = 0$. If $1 - \delta < \kappa \leq 1$, set

$$|\zeta_m|^2 = C^2 \quad (m \notin \mathcal{G}), \quad |\zeta_m|^2 = C^2 \frac{\kappa - 1 + \delta}{\delta} \quad (m \in \mathcal{G}).$$

This saturates the bad capacity and attains the positive branch of (8). For $\delta = 0$, necessarily $\sigma = 0$; for $\delta = 1$, necessarily $\sigma = 1$. These agree with the formula and complete the endpoint cases. $\square$

The equality mechanism is transparent. On the zero branch it is enough that all occupied energy fit on the bad set. On the positive branch, equality requires every positive-weight bad atom to be saturated; the remaining prescribed energy is then placed on the good set. No phase condition occurs.

Corollary 3.2 (Saturation-defect form). *Write $\epsilon := 1 - \kappa$. Then*

$$\sigma \geq \frac{(\delta - \epsilon)_+}{1 - \epsilon}. \quad (10)$$

If $\epsilon \leq (1 - \tau)\delta$ for some $0 \leq \tau \leq 1$, then

$$\sigma \geq \frac{\tau\delta}{1 - \epsilon} \geq \tau\delta. \quad (11)$$

The condition $\epsilon < \delta$ for a positive conclusion is sharp.

Thus “large total energy” is meaningful only relative to the capacity of the complement. A lower bound $E \gg C^2 W$ with an unspecified small constant need not select a small good set: the exact requirement is $\kappa > 1 - \delta$.

3.1 A local coefficient floor

The other natural certificate is local rather than global.

Theorem 3.3 (Sharp good-cell floor). *Suppose*

$$|\zeta_m| \geq \theta C \quad (m \in \mathcal{G}), \quad |\zeta_m| \leq C \quad (m \in \mathcal{M}), \quad 0 < \theta \leq 1. \quad (12)$$

If $E > 0$, then

$$\sigma \geq \frac{\theta^2 \delta}{\theta^2 \delta + 1 - \delta} = \mathsf{T}_{\theta^2}(\delta). \quad (13)$$

The bound is sharp.

Proof. Write $A = E(\mathcal{G})$ and $B = E(\mathcal{M} \setminus \mathcal{G})$. The hypotheses give

$$A \geq \theta^2 C^2 \delta W, \quad B \leq C^2(1 - \delta)W.$$

Because $A/(A + B)$ increases with A and decreases with B , the claimed bound follows. Equality is obtained by taking $|\zeta_m| = \theta C$ on the good set and $|\zeta_m| = C$ on the bad set, with the evident endpoint interpretations. $\square$

If both the actual value of κ and a good-cell floor are known, neither certificate should be discarded.

Theorem 3.4 (Optimal combined certificate). *Under (12), a positive capacity level κ is feasible only if $\kappa \geq \theta^2 \delta$. For every feasible $\kappa \in (0, 1] \cap [\theta^2 \delta, 1]$,*

$$\sigma \geq \frac{\max\{\theta^2 \delta, \kappa - (1 - \delta)\}}{\kappa}. \quad (14)$$

The bound is sharp for each (δ, θ, κ) .

Proof. The floor gives $E(\mathcal{G}) \geq \theta^2 \delta C^2 W$, while the bad-set capacity argument gives $E(\mathcal{G}) \geq (\kappa - (1 - \delta))C^2 W$. Dividing their maximum by $E = \kappa C^2 W$ proves the inequality.

If $\kappa \leq 1 - \delta + \theta^2 \delta$, put exactly the floor energy on the good set and distribute $(\kappa - \theta^2 \delta)C^2 W$ uniformly over the bad set. If $\kappa \geq 1 - \delta + \theta^2 \delta$, saturate the bad set and distribute the remaining $(\kappa - 1 + \delta)C^2 W$ uniformly over the good set. All squared amplitudes lie in $[0, C^2]$ and respect the floor, proving sharpness. The degenerate values $\delta = 0, 1$ follow directly. $\square$

After the floor is removed, the formula in [theorem 3.4](#) extends to $\theta = 0$ and reduces to the capacity law. If κ is not specified, minimizing over all feasible values recovers [theorem 3.3](#). If $|\zeta_m| = C$ on every atom, then $\kappa = 1$ and $\sigma = \delta$ exactly.

4 Sharp transport through source and cofactor weights

The previous section treated a fixed base measure. We now quantify how a good fraction changes when the atoms are reweighted.

4.1 An exact four-factor odds identity

The sharp inequalities below are consequences of an exact identity that also gives a necessary-and-sufficient average-data summary.

Theorem 4.1 (Exact row–shape–amplitude–occupancy factorization). *Let $q_m \geq 0$ be reference row weights, let $b_m \geq 0$ be a fixed base-energy multiplier, let $|\zeta_m| \leq C$, and let $N_m \geq 0$ be an occupancy multiplier. For $A \in \{\mathcal{G}, \mathcal{B}\}$, where $\mathcal{B} = \mathcal{M} \setminus \mathcal{G}$, set*

$$\begin{aligned} Q_A &:= \sum_{m \in A} q_m, & \bar{b}_A &:= \frac{\sum_{m \in A} q_m b_m}{Q_A}, \\ \vartheta_A &:= \frac{\sum_{m \in A} q_m b_m |\zeta_m|^2}{C^2 \sum_{m \in A} q_m b_m}, & \bar{N}_A &:= \frac{\sum_{m \in A} q_m b_m |\zeta_m|^2 N_m}{\sum_{m \in A} q_m b_m |\zeta_m|^2}. \end{aligned} \quad (15)$$

Put $\delta_0 = Q_{\mathcal{G}}/(Q_{\mathcal{G}} + Q_{\mathcal{B}})$, and define

$$F_A := \sum_{m \in A} q_m b_m |\zeta_m|^2 N_m, \quad \sigma_N := \frac{F_{\mathcal{G}}}{F_{\mathcal{G}} + F_{\mathcal{B}}}.$$

Assume, for both $A = \mathcal{G}, \mathcal{B}$, that Q_A , the base mass $\sum_A q_m b_m$, the amplitude mass $\sum_A q_m b_m |\zeta_m|^2$, and F_A are all positive. Then

$$\boxed{\frac{\sigma_N}{1 - \sigma_N} = \frac{\bar{N}_{\mathcal{G}} \vartheta_{\mathcal{G}} \bar{b}_{\mathcal{G}}}{\bar{N}_{\mathcal{B}} \vartheta_{\mathcal{B}} \bar{b}_{\mathcal{B}}} \frac{\delta_0}{1 - \delta_0}}. \quad (16)$$

The boundary cases are read directly from $F_{\mathcal{G}}$ and $F_{\mathcal{B}}$: the selected fraction is zero if only the bad energy is positive, one if only the good energy is positive, and undefined if both vanish.

Proof. For either A , the definitions telescope to

$$F_A = C^2 Q_A \bar{b}_A \vartheta_A \bar{N}_A.$$

Taking $F_{\mathcal{G}}/F_{\mathcal{B}}$ proves (16). $\square$

Thus $\sigma_N \geq s$, for $0 < s < 1$, is equivalent to the right side of (16) being at least $s/(1 - s)$. Pointwise coefficient floors and occupancy bounds are convenient sufficient conditions; the product of the good-to-bad weighted average ratios is an exact necessary-and-sufficient summary. All unit phases have disappeared before any estimate is made.

Taking $N_m \equiv 1$ and writing

$$\delta_b := \frac{\sum_{\mathcal{G}} q_m b_m}{\sum_{\mathcal{M}} q_m b_m}, \quad \sigma_b := \frac{\sum_{\mathcal{G}} q_m b_m |\zeta_m|^2}{\sum_{\mathcal{M}} q_m b_m |\zeta_m|^2},$$

gives the exact amplitude identity

$$\frac{\sigma_b}{1 - \sigma_b} = \frac{\vartheta_{\mathcal{G}}}{\vartheta_{\mathcal{B}}} \frac{\delta_b}{1 - \delta_b}, \quad (17)$$

Taking $b_m \equiv 1$ and $|\zeta_m| \equiv C$ isolates the exact energy-weighted occupancy ratio.

4.2 Pointwise sharp transport

Theorem 4.2 (Sharp occupancy transport). *Let $u_m \geq 0$ have positive total mass and let $\delta = \sum_{\mathcal{G}} u_m / \sum_{\mathcal{M}} u_m$. Let $v_m \geq 0$ satisfy*

$$v_m \geq v_- > 0 \quad (m \in \mathcal{G}), \quad v_m \leq v_+ < \infty \quad (m \notin \mathcal{G}), \quad v_+ > 0, \quad (18)$$

Assume $\sum_m u_m v_m > 0$. Then the good fraction after reweighting is

$$\delta_v := \frac{\sum_{m \in \mathcal{G}} u_m v_m}{\sum_{m \in \mathcal{M}} u_m v_m} \geq \mathsf{T}_{v_-/v_+}(\delta) = \frac{v_- \delta}{v_- \delta + v_+ (1 - \delta)}. \quad (19)$$

The inequality is sharp. If in addition $v_- \leq v_+$, $v_m \leq v_+$ on $\mathcal{G}$ and $v_m \geq v_-$ on its complement, the corresponding sharp upper bound is

$$\delta_v \leq \frac{v_+ \delta}{v_+ \delta + v_- (1 - \delta)}. \quad (20)$$

Proof. Put $A = \sum_{\mathcal{G}} u_m v_m$ and $B = \sum_{\mathcal{M} \setminus \mathcal{G}} u_m v_m$. Then $A \geq v_- \delta \sum u_m$ and $B \leq v_+ (1 - \delta) \sum u_m$. Monotonicity of $A/(A + B)$ gives the lower bound. Equality is attained by taking $v = v_-$ on $\mathcal{G}$ and $v = v_+$ on its complement. Reversing the two inequalities proves and sharpens the upper bound; equality there is attained by taking $v = v_+$ on $\mathcal{G}$ and $v = v_-$ on its complement. $\square$

Only a lower bound on good atoms and an upper bound on bad atoms are needed for the lower estimate. Very large good multipliers help rather than hurt. By contrast, a global upper bound with no good lower bound permits complete loss of occupancy.

The exact factorization also gives a weaker-hypothesis version. If $E_A = \sum_{m \in A} u_m |\zeta_m|^2 > 0$ and

$$\bar{N}_A = \frac{\sum_{m \in A} u_m |\zeta_m|^2 N_m}{E_A},$$

then, whenever both lifted masses are positive, the lifted odds are exactly the unlifted odds multiplied by $\bar{N}_{\mathcal{G}}/\bar{N}_{\mathcal{B}}$. If only the good lifted mass is positive the lifted fraction is one; if only the bad lifted mass is positive it is zero. Hence pointwise bounds $N_m \geq n_-$ on good atoms and $N_m \leq n_+$ on bad atoms are not necessary; they merely certify the corresponding average ratio.

4.3 Exact composition of the coverage ladder

Start with geometric weights q_m and good fraction δ_{geo} . Let a fixed source-shape multiplier have ratio $\lambda_{\text{src}} = a_-/a_+$, let the sampled-carrier multiplier have ratio $\lambda_{\text{car}} = b_-/b_+$, and let the cofactor-occupancy multiplier have ratio $\lambda_{\text{cf}} = n_-/n_+$, all in the sense of (18). Applying theorem 4.2 and lemma 2.1 gives

$$\delta_{\text{lift}} \geq \mathsf{T}_{\lambda_{\text{cf}}}(\mathsf{T}_{\lambda_{\text{car}}}(\mathsf{T}_{\lambda_{\text{src}}}(\delta_{\text{geo}}))) = \mathsf{T}_{\lambda_{\text{src}}\lambda_{\text{car}}\lambda_{\text{cf}}}(\delta_{\text{geo}}). \quad (21)$$

If also $|\zeta_m| \geq \theta C$ on the good set and $|\zeta_m| \leq C$ everywhere, one more application yields

$$\boxed{\sigma \geq \mathsf{T}_{\theta^2 \lambda_{\text{src}} \lambda_{\text{car}} \lambda_{\text{cf}}}(\delta_{\text{geo}}).} \quad (22)$$

Each stage, and the composition, is sharp in the unrestricted abstract finite-atom class: take every good multiplier at its lower endpoint and every bad multiplier at its upper endpoint. Literal arithmetic factors need not realize all these extremizers simultaneously.

Equation (22) is stronger than multiplying a good numerator by several lower bounds while repeatedly replacing the total denominator by a global upper bound. The odds coordinate keeps the exact denominator at every stage.

If a coefficient floor is unavailable, define κ relative to the already lifted base weights. Then theorem 3.1 gives instead

$$\sigma \geq \left(1 - \frac{1 - \delta_{\text{lift}}}{\kappa}\right)_+. \quad (23)$$

This route needs no pointwise lower bound, but near-saturation becomes more demanding when δ_{lift} is small.

4.4 A finite-cover alternative

Balance is sufficient, not logically necessary. A second route is to cover every active atom by finitely many compatible cells.

Proposition 4.3 (Selection from a finite compatible cover). *Suppose $E > 0$ and the support of the coefficient-weighted atomic energy is contained in $\mathcal{G}_1 \cup \dots \cup \mathcal{G}_R$, where every $\mathcal{G}_j$ has the required profile compatibility and R is independent of the scale. Then some cell satisfies*

$$\frac{E(\mathcal{G}_j)}{E} \geq \frac{1}{R}. \quad (24)$$

The constant $1/R$ is optimal.

Proof. Since the cover may overlap, $E \leq \sum_{j=1}^R E(\mathcal{G}_j)$. One summand is therefore at least E/R . Equality occurs for a partition into R cells of equal energy. $\square$

This proposition tolerates arbitrary source and cofactor concentration, but its premise is stronger geometrically: the compatible cells must cover the *actual active support*, not merely one open positive-volume subset of the ambient parameter box.

5 Specialization to the literal fixed-shift packet

We now insert the exact factors identified in TPC–53. This section does not alter the inherited coefficient class.

5.1 The physical structural weights

Fix the literal shift $h_0 \neq 0$, the opposite-row box $\mathcal{Q}$, the orbit interval I , and the direct residue class a used in the TPC–53 sampled carrier. Fix one complete source dyadic/residue cell $\mathcal{M}_X^{\text{full}}$ on this carrier slice and its compatible subset $\mathcal{M}_X^{\text{good}}$. Thus “full” below means complete only on the source side of the declared slice, not the whole TPC–48 residual packet. For a source row $m = d_m \ell_m$, with ℓ_m prime and d_m squarefree, remove the inherited factor $\zeta_m^{(1)}$ and define

$$u_m := |\mu(d_m)(\log \ell_m)\omega_D(d_m)\psi_L(\ell_m/L)|^2. \quad (25)$$

All fixed residue and smooth-shape restrictions are understood as part of the declared source cell. Let $\mathcal{I}_X$ be the retained pre-terminal set of pairs (m, r) , let $N_r(m) = \#\{r : (m, r) \in \mathcal{I}_X\}$, and let $\mathcal{B}_X^M(m) \geq 0$ be the sampled carrier norm of TPC–53. The pre-terminal carrier is identical for every retained r over the same source row; neither $\mu(r)\Lambda(pr)$ nor any terminal prime weight is included at this stage. The exact factorization is

$$b_m := u_m \mathcal{B}_X^M(m), \quad N_m := N_r(m), \quad w_m := b_m N_m. \quad (26)$$

Delete zero-weight rows by putting

$$\mathcal{A}_X := \{m \in \mathcal{M}_X^{\text{full}} : w_m > 0\}, \quad \mathcal{G}_X := \mathcal{A}_X \cap \mathcal{M}_X^{\text{good}}. \quad (27)$$

Consequently the actual-coefficient-weighted carrier energy is

$$\mathcal{C}_X = \sum_{m \in \mathcal{A}_X} w_m |\zeta_m^{(1)}|^2, \quad (28)$$

and, whenever $\mathcal{C}_X > 0$, the compatible carrier-energy fraction is

$$\sigma_X = \frac{\sum_{m \in \mathcal{G}_X} w_m |\zeta_m^{(1)}|^2}{\mathcal{C}_X}. \quad (29)$$

These formulas are simply the repeated orthogonal carrier normalization on (m, r) before the terminal prime map [4, 5]. The residual coordinate is not yet grouped.

TPC–53 proves, on a fixed interior compatible source/orbit box, a pointwise sampled comparison

$$\mathcal{P}_X^M(m) \geq \alpha_* \mathcal{B}_X^M(m), \quad \alpha_* > 0, \quad m \in \mathcal{G}_X, \quad (30)$$

for all large X . Define the exact unseparated physical-profile energies by

$$\mathcal{Q}_X := \sum_{m \in \mathcal{A}_X} u_m N_r(m) |\zeta_m^{(1)}|^2 \mathcal{P}_X^M(m), \quad (31)$$

$$\mathcal{Q}_{X,\text{good}} := \sum_{m \in \mathcal{G}_X} u_m N_r(m) |\zeta_m^{(1)}|^2 \mathcal{P}_X^M(m). \quad (32)$$

Therefore any lower bound for σ_X implies the pre-terminal physical-profile estimate

$$\mathcal{Q}_{X,\text{good}} \geq \alpha_* \sigma_X \mathcal{C}_X. \quad (33)$$

This implication is pointwise and has no local-to-global normalization gap.

5.2 Three coefficient regimes

Assume $\mathcal{C}_X > 0$, and let $C_X > 0$ be any certified upper bound such that $|\zeta_m^{(1)}| \leq C_X$ on the declared full source slice. With the active carrier weights w_m , define

$$\delta_{\text{cf},X} := \frac{\sum_{m \in \mathcal{G}_X} w_m}{\sum_{m \in \mathcal{A}_X} w_m}, \quad \kappa_X := \frac{\mathcal{C}_X}{C_X^2 \sum_{m \in \mathcal{A}_X} w_m}. \quad (34)$$

The universal sharp capacity theorem gives the valid physical specialization

$$\sigma_X \geq \left(1 - \frac{1 - \delta_{\text{cf},X}}{\kappa_X}\right)_+. \quad (35)$$

More precisely, [theorem 4.1](#) applies with reference row weight one on $\mathcal{M}_X^{\text{full}}$, fixed base multiplier $b_m = u_m \mathcal{B}_X^M(m)$, amplitude $\zeta_m^{(1)}$, and occupancy $N_m = N_r(m)$. It gives the exact identity

$$\frac{\sigma_X}{1 - \sigma_X} = \frac{\bar{N}_{\text{good}}}{\bar{N}_{\text{bad}}} \frac{\vartheta_{\text{good}}}{\vartheta_{\text{bad}}} \frac{\bar{b}_{\text{good}}}{\bar{b}_{\text{bad}}} \frac{|\mathcal{M}_X^{\text{good}}|}{|\mathcal{M}_X^{\text{full}} \setminus \mathcal{M}_X^{\text{good}}|}, \quad (36)$$

whenever all four factors are nondegenerate. This identity is the exact dictionary: raw row occupancy, carrier/source shape, coefficient magnitude, and cofactor occupancy are separate. The boundary cases are handled by the good and bad carrier energies, not by assigning infinite ratios.

This formula separates three cases.

- (a) For the full inherited class, only $|\zeta_m^{(1)}| \leq C_X$ is known. Provided $\mathcal{A}_X \setminus \mathcal{G}_X \neq \emptyset$, the bounded class permits a coefficient that vanishes on every good row and has $\sigma_X = 0$. If $\mathcal{A}_X = \mathcal{G}_X$, instead $\delta_{\text{cf},X} = \sigma_X = 1$ for every positive carrier energy.
- (b) Suppose a quantitative provenance calculation on a fixed direct residue cell verifies

$$0 < c_X \leq |\zeta_m^{(1)}| \quad (m \in \mathcal{G}_X), \quad |\zeta_m^{(1)}| \leq C_X \quad (m \in \mathcal{A}_X).$$

Then

$$\sigma_X \geq \mathbb{T}_{(c_X/C_X)^2}(\delta_{\text{cf},X}).$$

Abstract diagonal unit phases attached to these coefficients do not change the conclusion. The Mellin factors in the literal dictionary remain in the provenance measure or layer phase; no favorable Mellin layer is selected and no signed-reassembly lower bound is inferred. The displayed floor is an extra checkable quantitative hypothesis, not a consequence of qualitative nonvanishing and not a theorem about every bounded inherited coefficient [\[1, 3\]](#).

- (c) If $|\zeta_m^{(1)}| = C_X$ on the declared active packet, then $\sigma_X = \delta_{\text{cf},X}$ exactly. Even here, source-row density alone does not control $\delta_{\text{cf},X}$ if cofactor multiplicities concentrate on bad rows.

Alternatively, without a coefficient floor, the single scalar estimate

$$\kappa_X > 1 - \delta_{\text{cf},X} \quad (37)$$

forces positive selection. The threshold cannot be weakened using only the displayed data.

5.3 What geometric information already supplies

Assume explicitly that the compatible source box contains a fixed positive-volume subrectangle on which the declared unary source shapes ω_D and ψ_L are bounded away from zero. Under this compatibility hypothesis, the fixed-modulus prime-squarefree count and weighted asymptotic of TPC-50 give

$$\delta_{\text{src},X} := \frac{\sum_{m \in \mathcal{M}_X^{\text{good}} u_m}{\sum_{m \in \mathcal{M}_X^{\text{full}} u_m} \gg 1. \quad (38)$$

This is a source-base statement, not yet a carrier statement [3, 5].

On the fixed sampled carrier slice, the carrier lower bound on the good box, the global fixed-shape upper bound, and uniform mask count can be recorded as

$$B_- S_X \leq \mathcal{B}_X^M(m) \quad (m \in \mathcal{M}_X^{\text{good}}), \quad \mathcal{B}_X^M(m) \leq B_+ S_X \quad (m \in \mathcal{M}_X^{\text{full}}), \quad (39)$$

with $S_X > 0$ and fixed $0 < B_- \leq B_+ < \infty$. These are the sampled two-sided carrier data required from the declared TPC-53 box; if they are not verified for a larger full source slice, they must be retained as an explicit hypothesis. Define

$$\delta_{\text{car},X} := \frac{\sum_{m \in \mathcal{M}_X^{\text{good}} u_m \mathcal{B}_X^M(m)}{\sum_{m \in \mathcal{M}_X^{\text{full}} u_m \mathcal{B}_X^M(m)}. \quad (40)$$

Then

$$\delta_{\text{car},X} \geq \mathsf{T}_{B_-/B_+}(\delta_{\text{src},X}) \gg 1. \quad (41)$$

Neither statement yet gives the cofactor-lifted fraction $\delta_{\text{cf},X}$.

For example, suppose one additionally proves, for some scalar $R_X^{\text{occ}} > 0$, the balanced-occupancy condition

$$n_- R_X^{\text{occ}} \leq N_m \leq n_+ R_X^{\text{occ}} \quad \text{on every positive-carrier full source row,} \quad 0 < n_- \leq n_+ < \infty. \quad (42)$$

Then the sharp transport theorem gives

$$\delta_{\text{cf},X} \geq \frac{n_- \delta_{\text{car},X}}{n_- \delta_{\text{car},X} + n_+ (1 - \delta_{\text{car},X})}. \quad (43)$$

Together with a fixed nonzero coefficient floor, this yields $\sigma_X \gg 1$ and hence (33). The inherited supported-pair interface permits arbitrary finite cofactor multiplicity and does not prove (42).

There are therefore three honest ways to finish this selection subgate: prove direct actual occupancy $\delta_{\text{cf},X} \gg 1$; prove a product-like estimate such as (42); or construct a fixed finite compatible cover of every nonzero active carrier and apply [proposition 4.3](#). TPC-53 proves openness of one active compatible set, not the third global cover.

Remark 5.1 (Carrier versus cofactor balance). The carrier factor in the literal dictionary is independent of the real Mellin-layer parameter and is uniformly two-sided on the declared fixed sampled cell. This can reduce (42) to a statement about $N_r(m)$ after all relevant cell-uniformity constants are recorded. It does not itself establish that statement.

6 Phase blindness and the residual-grouping barrier

The selection theory is deliberately placed before the terminal map. Two elementary invariances show why this order cannot be reversed for free.

6.1 Diagonal phases cannot repair magnitude coverage

Proposition 6.1 (Pre-terminal phase blindness). *For arbitrary real phases ϑ_m , replace ζ_m by $e^{i\vartheta_m}\zeta_m$. Then $E(S)$, δ , κ , and σ are unchanged for every $S \subseteq \mathcal{M}$. The same holds after orthogonal repetition over retained labels (m, r) .*

Proof. Every displayed quantity is a weighted sum of $|\zeta_m|^2$, and $|e^{i\vartheta_m}\zeta_m| = |\zeta_m|$. Repetition over orthogonal coordinates only repeats the same diagonal squared modulus. $\square$

Thus equidistribution, randomness, or oscillation of phases alone cannot move pre-terminal energy onto a good cell. Such information can become relevant only when amplitudes are summed coherently later.

6.2 A two-atom kernel after grouping

Let $\mathcal{U}$ be a finite set of admissible post-prime input labels and let the residual grouping map have the fiber form

$$(Tz)_y = \sum_{u: q(u)=y} c_u z_u, \quad (44)$$

In the literal slice, one may take

$$u = (m, r, j, \gamma), \quad q(u) = (\gamma, d(m)r, j),$$

where the terminal prime p_u is admissible and satisfies $p_u r - mj = h_0$; the coefficient c_u includes $\mu(r)\Lambda(p_u r)$ and the already formed prime amplitude. Thus equality of $d(m)r$ alone is not enough for a physical collision: the other output coordinates and terminal constraint must also agree. This notation isolates only the linear-algebra issue and does not replace the literal terminal formula of TPC-53.

Theorem 6.2 (Two-atom collision obstruction). *Suppose one output fiber contains distinct labels u_1, u_2 with $c_{u_1}c_{u_2} \neq 0$. Then T has a nonzero vector supported on that fiber in its kernel. Consequently there is no constant $c > 0$ such that*

$$\|Tz\|_2^2 \geq c \sum_{u \in \mathcal{U}} |z_u|^2 \quad (45)$$

for all coefficient arrays supported on any selected subspace containing both coordinate directions.

Proof. Take

$$z_{u_1} = c_{u_2}, \quad z_{u_2} = -c_{u_1}, \quad z_u = 0 \quad (u \neq u_1, u_2).$$

Then the only potentially nonzero output is $c_{u_1}c_{u_2} - c_{u_2}c_{u_1} = 0$, while $z \neq 0$. Scaling the vector also meets any prescribed common upper bound on its coordinates. $\square$

At a fixed source m , the map $r \mapsto d(m)r$ is injective, so the literal pre-terminal r -square is exact. Collisions, if present, can arise only across different sources. TPC-53 displays the actual order: multiply by the terminal weight, form the coherent prime sum, and only then group equal values of $s = d(m)r$. Expanding the final squared norm produces cross terms absent from (28).

Corollary 6.3 (Logical boundary of energy selection). *If a selected compatible coordinate subspace contains both directions in such a nonzero collision fiber, it has a kernel vector with compatible input-energy fraction one. Hence no uniform grouped lower bound holds on that subspace. A coverage statement alone gives no lower bound until one checks how its selected support meets the physical collision fibers. To pass the boundary one must prove an estimate for the distinguished arithmetic array—for example a coherent-prime lower square followed by grouping stability—or restrict to a collision-free set of labels.*

The theorem is a no-go for an *automatic implication*, not evidence that the physical coefficients actually choose a cancelling vector. It pinpoints the next missing estimate without assigning adversarial freedom to a fixed arithmetic sequence.

7 Endpoint ledger, conclusions, and next gate

7.1 Polynomial selection cost

The odds formula makes the endpoint budget explicit. Suppose, along a sequence of scales,

$$\begin{aligned}\delta_{\text{geo}} &= X^{-\nu+o(1)}, & \lambda_{\text{src}} &= X^{-\omega_{\text{src}}+o(1)}, \\ \lambda_{\text{car}} &= X^{-\omega_{\text{car}}+o(1)}, & \lambda_{\text{cf}} &= X^{-\omega_{\text{cf}}+o(1)}, \\ \theta &= X^{-\chi+o(1)},\end{aligned}\tag{46}$$

with nonnegative exponents. Put

$$\mathbf{a}_X := \theta^2 \lambda_{\text{src}} \lambda_{\text{car}} \lambda_{\text{cf}}.$$

The elementary bound

$$\mathsf{T}_{\mathbf{a}_X}(\delta_{\text{geo}}) \geq \frac{\mathbf{a}_X \delta_{\text{geo}}}{1 + \mathbf{a}_X}$$

and (22) give

$$\sigma_X \geq X^{-(\nu+\omega_{\text{src}}+\omega_{\text{car}}+\omega_{\text{cf}}+2\chi)+o(1)}.\tag{47}$$

Thus the displayed sum is a sharp worst-case upper bound for the polynomial selection cost; the actual odds can be better. The factor two on χ is forced by squared energy. No phase exponent appears.

The inherited TPC-52 ledger, with fixed cell scale and fixed positive profile overlap, allows at most the nominal selection loss

$$\boxed{\nu + \omega_{\text{src}} + \omega_{\text{car}} + \omega_{\text{cf}} + 2\chi \leq \frac{1}{400}.}\tag{48}$$

At equality, an error already of size $X^{-1/400+o(1)}$ need not be absorbed without a constant-level comparison or an additional saving; strict slack is the safe closure condition. The fixed compatible boxes, fixed nonzero shapes, fixed sampled-carrier ratio, and fixed overlap α_* used in the declared TPC-53 cell have zero positive polynomial cost. Under fixed-ratio coefficient and occupancy bounds, the selection cost is therefore also zero.

If instead one uses the capacity route and such an exponent exists along the chosen subsequence, define ς by

$$\left(1 - \frac{1 - \delta_{\text{cf},X}}{\kappa_X}\right)_+ = X^{-\varsigma+o(1)}.$$

The value ς replaces the sum in (48). If the left side is zero infinitely often, or is smaller than X^{-A} for every fixed $A > 0$, this route fails the endpoint gate.

A shrinking profile threshold would introduce its own exponent. The existing sampled approximation supports the fixed interior $\alpha_* > 0$ regime; it should not be silently used as a relative-error theorem at a polynomially vanishing threshold.

7.2 Claim ledger

The exact status is as follows.

Statement	Status
Capacity threshold and extremizers	Unconditional finite-atom theorem (L0).
Odds transport and composition	Unconditional sharp change-of-weight theorem (L0).
Literal fixed- h_0 carrier normalization	Exact specialization of the TPC-53 pre-terminal carrier dictionary on the declared slice (L1 interface).
Compatible geometric/source box	Previously certified for fixed boxes; not full actual-coefficient carrier-energy coverage.
Positive declared-slice σ_X	Conditional on actual occupancy, balanced multiplicity plus a coefficient floor, capacity saturation, or a finite compatible cover.
Post-grouping lower bound	Not proved; impossible uniformly on a selected subspace containing both directions of a two-atom collision.
Parity or twin-prime consequence	Not claimed.

In particular, the following implications remain forbidden: geometric density does not imply actual-coefficient carrier energy; a unit-modulus source factor does not imply full lifted coverage without occupancy control; arbitrary cofactor multiplicity inside one selected packet does not imply that the packet captures a fixed fraction of the full carrier; and a pre-terminal lower bound does not commute with coherent prime summation or residual grouping.

7.3 The next gate

The correct order of further work is now forced by the maps:

1. select actual retained (m, r) -fibers by one of the sharp certificates above;
2. prove a two-sided or lower coherent-prime square while the source and cofactor labels remain separate;
3. control collisions in the grouping $s = d(m)r$;
4. localize the resulting estimate to the literal fixed shift h_0 and compare the residual Gram with the native affine Gram; and
5. reconnect the controlled packet to the original ultra-tail defect.

The immediate next paper takes the first step after selection: it keeps the residual grouping explicit, rewrites fibers by $s = d(m)r$, and asks which exact divisor/cofactor structure can prevent the two-atom kernel from being the relevant arithmetic behavior. The present paper has made that question quantitative, but has not answered it.

Conclusion. Given declared structural, capacity, and occupancy data, the abstract finite-atom selection problem left by the literal factor dictionary has a complete sharp answer. A positive lifted structural fraction is selected if it also has a local coefficient floor; alternatively, global coefficient energy may saturate capacity beyond the complement threshold, or a finite compatible cover may capture all active carrier energy. Source, carrier, and cofactor distortions act multiplicatively on odds and add at the exponent level. These facts are exact before terminal grouping. The same exactness also shows why the result stops there: phases are invisible before the sum, and collisions can cancel after it.

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